You have your sample application, a class to test, and a test project, so it's about time you wrote some tests.

35.5 Adding Fact and Theory unit tests

When I write unit tests, I usually target one of three paths through the method under test:

- *The happy path*—Where typical arguments with expected values are provided
- *The error path*—Where the arguments passed are invalid and tested for
- *Edge cases*—Where the provided arguments are right on the edge of expected values

I realize that this is a broad classification, but it helps me think about the various scenarios I need to consider.

TIP A completely different approach to testing is property-based testing. This fascinating approach is common in functional programming communities, like F#. You can find a great introduction by Scott Wlaschin in his blog post series "The 'Property Based Testing' Series" at http://mng.bz/o1eZ. That post uses F#, but it is still highly accessible even if you're new to the language.

Let's start with the happy path, writing a unit test that verifies that the `ConvertToGbp()` method is working as expected with typical input values, as shown in the following listing.

Listing 35.4 Unit test for `ConvertToGbp` using expected arguments

```
[Fact]                                                    ❶
public void ConvertToGbp_ConvertsCorrectly()              ❷
{
```